Lab Report

Study of Josephson-Junction and Flux-Quantization Properties via SQUID

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Abstract

The current–voltage (V–I) and voltage–flux (V–Φ) characteristics of a high-temperature DC SQUID magnetometer were recorded at 77 K using a two-channel digital oscilloscope. From the V–I trace, the junction critical current was found to be $I_c = 69.5$ µA and the normal-state resistance $R_N = 4.47\ \Omega$ (per junction), giving a characteristic voltage $I_c R_N = 310$ µV. From the V–Φ trace, the peak-to-peak modulation depth was $\Delta V = 23.0$ µV with a period corresponding to one flux quantum $\Phi_0 = 2.07 \times 10^{-15}$ Wb. Using the chip inductance of $L \approx 60$ pH, the dimensionless modulation parameter evaluates to $\beta_L = 4.03$ from the Eqn. ($\beta_L = 2I_c L/\Phi_0$) and $\beta_L = 16.2$ from the zero-temperature Eqn. ($\beta_L = 4I_c R_N/\pi\Delta V - 1$). This disagreement occurs because the basic formulas were developed for traditional low-temperature SQUIDs and do not account for thermal noise. Because Mr. SQUID® operates at the relatively high temperature of liquid. Applying the thermally corrected expression reduces this to $\beta_L = 8.70$, roughly halfway toward the inductance-based value.

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1 Objective

A direct-current Superconducting QUantum Interference Device (dc SQUID) is a ring of superconductor interrupted by two Josephson junctions connected in parallel. Two phenomena make the device useful as a sensitive magnetometer: (i) DC Josephson effect, by which a resistanceless (super-)current can flow through each junction up to a critical value I_c; and (ii) flux quantization, which forces the magnetic flux threading the superconducting loop to change in discrete steps of the flux quantum $\Phi_0 = h/2e = 2.07 \times 10^{-15}$ Wb. Together these give the SQUID a critical current, and hence a terminal voltage at fixed bias, that is a periodic function of applied magnetic flux with period Φ_0.

The aims of this experiment were:

- To measure V–I characteristic, extract junction critical current I_c and normal-state resistance R_N.
- To measure V–Φ characteristic, and extract peak-to-peak voltage modulation ΔV and flux period.
- To evaluate the modulation parameter β_L by two independent methods and compare the results.

2 Introduction

2.1 The Josephson Junction

A Josephson junction consists of two superconducting regions that are weakly coupled to one another, for example by a thin insulating gap, a physical constriction, or a crystallographic grain boundary. Despite the weak link, the junction behaves as a single superconductor: it supports a resistanceless, zero-voltage current carried by Cooper pairs tunnelling between the two regions. This current can flow only up to a maximum value, the **critical current** I_c. Once the total current through the junction exceeds I_c, the junction switches into a resistive state and a voltage develops across it.

2.2 Why Two Josephson Junctions are Used

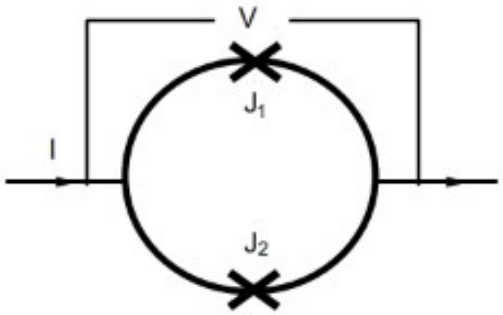


Figure 1: Schematic representation of a DC SQUID

The DC SQUID is constructed from **two** Josephson junctions connected in parallel on a single closed superconducting loop. This arrangement is chosen to exploit macroscopic quantum interference: the superconducting wavefunctions at the two junctions interfere with one another, in direct analogy with light waves passing through a double-slit optical experiment. As a consequence, the critical current of the loop as a whole is not constant but is highly sensitive to the magnetic flux threading the loop, oscillating periodically between a maximum and a minimum value as the applied flux is varied. A single-junction device (the rf SQUID) can exhibit a related effect, but only the two-junction parallel geometry produces this interference-based modulation of critical current with magnetic flux.

2.3 Faraday's Law and Lenz's Law in the Superconducting Loop

The response of SQUID loop to a magnetic field is governed by two classical electromagnetic principles:

- **Faraday's Law of Induction** states that whenever the magnetic flux threading a closed loop changes, an electromotive force is induced, driving a current in the loop.
- **Lenz's Law** determines the direction of this induced current: it flows so as to generate a magnetic field equal in magnitude and exactly opposite in direction to the field that produced it, opposing the change in flux.

In an ordinary resistive ring, the induced current would decay away rapidly. In a superconducting ring, however, zero resistance ensures that the induced **screening current** does not decay. It flows indefinitely, continuously opposing any attempted change in applied flux and thereby enforcing the requirement that the total flux enclosed by the loop remain an integer multiple of the flux quantum, $\Phi_0 = h/2e$.

2.4 Working Principle of the DC SQUID

The DC SQUID functions as an ultra-sensitive magnetic flux-to-voltage transducer. Its operating cycle:

1. A constant DC bias current (slightly greater than the critical current I_c) is driven through the SQUID so that the device always operates in a resistive state with a measurable voltage across it.
2. When an external magnetic field is applied, the loop generates a circulating screening current I_s (by Faraday's and Lenz's Laws) to oppose the applied field and keep the net enclosed flux pinned to an integral number of flux quanta.
3. This screening current superimposes on the bias current flowing through the junctions. Since each junction can carry only a fixed total current before becoming resistive, the additional screening current effectively lowers the SQUID's usable critical current.
4. As the applied flux increases, I_s grows correspondingly, reaching its maximum value when the applied flux is offset from an integer multiple of Φ_0 by exactly one half of a flux quantum. At this point, the junctions momentarily and briefly become normal, allowing one additional quantum of flux to enter the loop; the screening current then reverses direction and the cycle restarts.
5. Because this process repeats identically every time one flux quantum passes through the loop, the screening current, the critical current $I_c(\Phi)$, and the resulting output voltage V across the SQUID all oscillate periodically with the applied flux, with a period of exactly one flux quantum, Φ_0.

This V–Φ relationship allows the SQUID to convert an unknown magnetic flux into an easily measurable voltage signal.

2.5 Changing Flux in the SQUID Instrument

In the Mr. SQUID® instrument, magnetic flux applied to the chip can be controlled in several ways:

- **FLUX OFFSET control:** A front-panel knob that drives a variable DC current through an "internal" feedback coil fabricated directly on the SQUID chip. The resulting local magnetic field couples directly into the SQUID loop, allowing manual adjustment of the applied flux.
- **SWEEP OUTPUT control:** Automates this process by driving an oscillating (triangle-wave) current through the same feedback coil, continuously sweeping the flux back and forth so that the full voltage-flux (V-Φ) characteristic can be observed on an oscilloscope.
- **External sources:** Flux can also be altered by bringing a small permanent magnet near the probe, or by physically rotating the probe with respect to the Earth's magnetic field.

2.6 BNC Cables and Their Purpose

BNC (Bayonet Neill–Concelman) coaxial cables are connectors used to transmit electronic signals with minimal interference and signal loss. Here, they connect `VOLTAGE` and `CURRENT` outputs on the front panel of the electronics control box to vertical and horizontal inputs of oscilloscope. This connection scheme allows V-I and V-Φ characteristics of the SQUID to be displayed and monitored in real time.

3 Theory

The modulation parameter of a SQUID is defined as

$$\beta_L = \frac{2I_c L}{\Phi_0} \qquad \text{(Eqn. 3-1)}$$

where L is the self-inductance of the SQUID loop. Because L is difficult to measure directly, β_L can instead be estimated from the characteristic voltage $I_c R_N$ and the observed modulation depth ΔV,

$$\beta_L = \frac{4}{\pi}\frac{I_c R_N}{\Delta V} - 1 \qquad \text{(Eqn. 3-2)}$$

Eqn. 3-2 is valid only when the two junctions are identical and thermal noise is negligible; Mr. SQUID® operates at 77 K, where $k_B T$ is not small compared with Josephson coupling energy, so the equations are not expected to agree. At nonzero temperature, thermal fluctuations round the V–Φ modulation and reduce ΔV below the value Eqn. 3-2 would predict from $I_c R_N$ alone. The corrected version is

$$\beta_L = \frac{4I_c R_N}{\pi \Delta V}\left(1 - 3.57\sqrt{\frac{k_B T L}{\Phi_0^2}}\right) - 1 \qquad \text{(Eqn. 3-5)}$$

where $k_B = 1.381 \times 10^{-23}$ J/K is Boltzmann's constant and $T = 77$ K is the bath temperature. The correction factor in parentheses is always ≤ 1, so Eqn. 3-5 pulls the modulation-depth estimate of β_L down.

For an N-turn modulation coil on the SQUID washer, the mutual inductance between coil and loop is

$$M = N\big[0.5(L_h + L_k + L_{sl}) + L_j\big] \qquad \text{(Eqn. 3-3)}$$

and can be measured directly from the period of the V–Φ curve using

$$M = \frac{\Phi_0}{\Delta I} \qquad \text{(Eqn. 3-4)}$$

where ΔI is the modulation-coil current needed to advance the applied flux by one quantum Φ_0. For the washer geometry of Mr. SQUID® ($d = 24$ µm hole, 33 µm slit) total loop inductance is $L \approx 60$ pH.

The critical current and normal-state resistance are read from the two asymptotic (resistive) branches of the V–I curve: the two branches are each extrapolated linearly back to $V = 0$, and the separation of the two zero-crossings equals $4I_c$ (the width of the flat, zero-voltage plateau, corrected for the fact that the bias current splits between the two junctions and sweeps symmetrically about zero). The slope of each branch equals $R_N/2$ for the SQUID as a whole, because the two junctions are in parallel.

4 Experimental Setup and Procedure

4.1 Apparatus

- Mr. SQUID® electronic control box (STAR Cryoelectronics), with internal voltage gain 10,000 and current-monitor resistor 10 kΩ.
- Mr. SQUID® cryogenic probe containing a YBCO thin-film dc SQUID chip with internal/external modulation coils.
- Liquid-nitrogen dewar (77 K bath).
- Tektronix TDS2022C two-channel digital oscilloscope, used to record the X (current-proportional) and Y (voltage-proportional) outputs of the control box as CSV waveform files.

4.2 Procedure

1. Cooled the probe in liquid nitrogen with the control box disconnected, to minimize flux trapping.
2. Connected the probe, set the function switch to V–I, and adjusted the flux-bias control to maximize the width of the critical-current plateau.
3. Recorded one full current sweep on CH1 (X, current) and CH2 (Y, voltage) of the oscilloscope.
4. Switched the function switch to V–Φ, biased the SQUID at the knee of the V–I curve, and increased the amplitude control to sweep several periods of flux through the loop.
5. Recorded the modulation-coil current on CH1 and the resulting SQUID voltage on CH2.

4.3 Sources of Uncertainty

The dominant uncertainty in I_c and R_N comes from the linear extrapolation of the resistive branches of the V–I curve, which is somewhat sensitive to the current range chosen for the fit (branch curvature near the knee is thermally rounded). The dominant uncertainty in the V–Φ period is the single-sample peak/trough localization at a 2500-point, 20 µs-sample-interval digitization; four consecutive periods were averaged to reduce this. Oscilloscope vertical resolution (8-bit) and residual 60 Hz / RF pickup are additional minor noise sources, visible as a point-to-point scatter.

5 Observations and Data

Each file contains 2500 samples at a 40 µs sample interval (100 ms total record).

Table 1: Oscilloscope channel assignment and settings for the four recorded traces.

Mode	Channel	Quantity	Vertical scale
V–I	CH1 (X)	Current monitor	5.0 V/div
V–I	CH2 (Y)	Voltage monitor	5.0 V/div
V–Φ	CH1 (X)	Modulation-coil current	5.0 V/div
V–Φ	CH2 (Y)	SQUID voltage	0.5 V/div

Both the current-monitor (X) and voltage-monitor (Y) outputs of the control box are calibrated by the same factor of 10,000: the X output is the voltage across an internal 10 kΩ resistor ($I_{\text{actual}} = V_X/10\,\text{k}\Omega$), and the Y output is the SQUID voltage amplified by a factor of 10,000 ($V_{\text{actual}} = V_Y/10{,}000$).

6 Analysis and Results

6.1 Data Reduction

The raw (t, V) waveforms for CH1 and CH2 were paired sample-by-sample (identical sample clock) to form the parametric curves $V_Y(V_X)$. For the V–I curve, the data were sorted by V_X and bin-averaged (300 bins) to remove the small hysteresis between the rising and falling legs of the triangle-wave sweep. For the V–Φ curve, a single monotonic (falling) leg of the triangle sweep, $t \in [-27.7, -20.6]$ ms, was isolated to avoid overlapping multiple sweep directions on the same plot.

6.2 Critical Current and Normal-State Resistance (V–I curve)

The two resistive (approximately linear) branches of the bin-averaged V–I curve were fit over the ranges $V_X \in (10, 24)$ V and $V_X \in (-24, -10)$ V (well away from the knee, short of the amplifier saturation at $|V_X| \gtrsim 25$ V):

$$V_Y = (1.1219 \pm 0.0005)\, V_X - 0.0774 \qquad (V_X > 0)$$

$$V_Y = (1.1126 \pm 0.0005)\, V_X + 3.0145 \qquad (V_X < 0)$$

Setting $V_Y = 0$ in each line gives the zero-crossing (critical-current) intercepts

$$V_X^+ = 0.0690 \text{ V}, \qquad V_X^- = -2.7095 \text{ V},$$

so flat-plateau width is $\Delta V_X = V_X^+ - V_X^- = 2.7785$ V. Converting through 10 kΩ current-monitor resistor,

$$I_{\text{flat}} = \frac{\Delta V_X}{10\,\text{k}\Omega} = \frac{2.7785 \text{ V}}{10{,}000\ \Omega} = 277.85\ \mu\text{A}.$$

This flat width equals $4I_c$ (current through *both* junctions, swept symmetrically about zero), so the critical current of a single junction is

$$I_c = \frac{I_{\text{flat}}}{4} = \frac{277.85\ \mu\text{A}}{4} = \boxed{69.46\ \mu\text{A}}. \tag{1}$$

The two-junction (SQUID) critical current is $2I_c = 138.93$ µA, within the typical range of 5–150 µA.

Average slope of the two branches, $\overline{m} = (1.1219 + 1.1126)/2 = 1.1172$, is in units of V_Y/V_X; since both axes carry the same calibration factor, this ratio is numerically equal to a resistance in ohms. Because the two junctions are in parallel, this measured slope equals half the normal-state resistance of a single junction's parallel combination (i.e., $R_N^{\text{SQUID}}/1$, which is identified with $R_N/2$ of one junction):

$$R_N/2 = \overline{m} = 1.1172\ \Omega \quad \Rightarrow \quad R_N^{\text{SQUID}} = 2\overline{m} = 2.2345\ \Omega,$$

$$R_N \text{ (single junction)} = 2 \times R_N^{\text{SQUID}} = 4 \times \overline{m} = \boxed{4.469\ \Omega}. \tag{2}$$

The characteristic (Josephson) voltage of the junction is then

$$I_c R_N = (69.46\ \mu\text{A})(4.469\ \Omega) = \boxed{310.4\ \mu\text{V}}. \tag{3}$$

This is above the theoretical range of 10–100 µV for this device at 77 K; no correction has been applied, and the value is carried forward as measured (see Discussion).

6.3 Modulation Depth and Flux Period (V–Φ curve)

We obtain four clearly resolved oscillations across one monotonic leg of the modulation-coil sweep. The X-axis (modulation-coil current monitor) positions of the four successive maxima and minima are:

$$V_X^{\max} = \{1.8,\ 12.2,\ 21.8,\ 31.0\}\ \text{V}, \qquad V_X^{\min} = \{-3.0,\ 7.0,\ 15.8,\ 26.2\}\ \text{V}.$$

The successive spacings (one period = one flux quantum Φ_0 each) are

$$\Delta V_X = \{10.4,\ 9.6,\ 9.2\}\ \text{(peaks)}, \qquad \{10.0,\ 8.8,\ 10.4\}\ \text{(troughs)}\quad [\text{V}],$$

with mean $\overline{\Delta V_X} = 9.733$ V and sample standard deviation 0.653 V ($n = 6$). Converting through the 10 kΩ current-monitor resistor (current gain 10,000 V/A),

$$\Delta I = \frac{\overline{\Delta V_X}}{10{,}000\ \text{V/A}} = \frac{9.733\ \text{V}}{10{,}000\ \text{V/A}} = \boxed{973.3 \pm 65.3\ \mu\text{A}} \tag{4}$$

per flux quantum. By Eqn. 3-4, the coil-to-SQUID mutual inductance is

$$M = \frac{\Phi_0}{\Delta I} = \frac{2.07 \times 10^{-15}\ \text{Wb}}{973.3 \times 10^{-6}\ \text{A}} = \boxed{2.13 \pm 0.14\ \text{pH}}. \tag{5}$$

(M is a property of the external-coil/SQUID coupling, and is not required for any β_L calculation)

The mean raw (Y-axis) voltage of the four peaks and four troughs is

$$\overline{V_Y^{\max}} = -5.045\ \text{V}, \qquad \overline{V_Y^{\min}} = -5.275\ \text{V},$$

giving a raw peak-to-peak swing of $\overline{V_Y^{\max}} - \overline{V_Y^{\min}} = 0.230$ V. Dividing by the voltage-channel amplification,

$$\Delta V = \frac{0.230\ \text{V}}{10{,}000} = \boxed{23.0\ \mu\text{V}}, \tag{6}$$

which sits squarely inside the typical range of 10–30 µV.

6.4 Modulation Parameter β_L

Using Eqn. 3-1 with $L = 60$ pH and the measured $I_c = 69.46$ µA,

$$\beta_L^{(3\text{-}1)} = \frac{2I_cL}{\Phi_0} = \frac{2(69.46 \times 10^{-6}\ \text{A})(60 \times 10^{-12}\ \text{H})}{2.07 \times 10^{-15}\ \text{Wb}} = \boxed{4.03}. \tag{7}$$

Using Eqn. 3-2 with the measured $I_cR_N = 310.4$ µV and $\Delta V = 23.0$ µV,

$$\beta_L^{(3\text{-}2)} = \frac{4}{\pi}\frac{I_cR_N}{\Delta V} - 1 = \frac{4}{\pi}\left(\frac{310.4\ \mu\text{V}}{23.0\ \mu\text{V}}\right) - 1 = \frac{4}{\pi}(13.496) - 1 = 17.18 - 1 = \boxed{16.18}. \tag{8}$$

Using Eqn. 3-5 with $T = 77$ K and $L = 60$ pH, the correction term is first evaluated on its own:

$$\sqrt{\frac{k_BTL}{\Phi_0^2}} = \sqrt{\frac{(1.381 \times 10^{-23}\ \text{J/K})(77\ \text{K})(60 \times 10^{-12}\ \text{H})}{(2.07 \times 10^{-15}\ \text{Wb})^2}} = \sqrt{1.489 \times 10^{-2}} = 0.1220,$$

$$1 - 3.57\sqrt{\frac{k_BTL}{\Phi_0^2}} = 1 - 3.57(0.1220) = 1 - 0.4356 = 0.5644.$$

Applying this correction factor to the same prefactor used for Eqn. 3-2,

$$\beta_L^{(3\text{-}5)} = \frac{4I_cR_N}{\pi\Delta V}\left(1 - 3.57\sqrt{\frac{k_BTL}{\Phi_0^2}}\right) - 1 = (17.18)(0.5643) - 1 = 9.70 - 1 = \boxed{8.70}. \tag{9}$$

As expected, the thermal correction pulls the modulation-depth estimate down from 16.18 to 8.70 – roughly halfway to the inductance-based estimate $\beta_L^{(3\text{-}1)} = 4.03$, without fully closing the gap.

Table 2: Summary of measured SQUID parameters

Quantity	Symbol	Measured value
Critical current (junction)	I_c	69.46 µA
Normal-state resistance (junction)	R_N	4.469 Ω
Characteristic voltage	I_cR_N	310.4 µV
Modulation depth (peak-to-peak)	ΔV	23.0 µV
Mutual inductance (modulation coil)	M	2.13 pH
SQUID loop inductance	L	60 pH
β_L from inductance (Eqn. 3-1)	$\beta_L^{(3\text{-}1)}$	4.03
β_L from modulation depth (Eqn. 3-2)	$\beta_L^{(3\text{-}2)}$	16.18
β_L from modulation depth, thermally corrected (Eqn. 3-5)	$\beta_L^{(3\text{-}5)}$	8.70

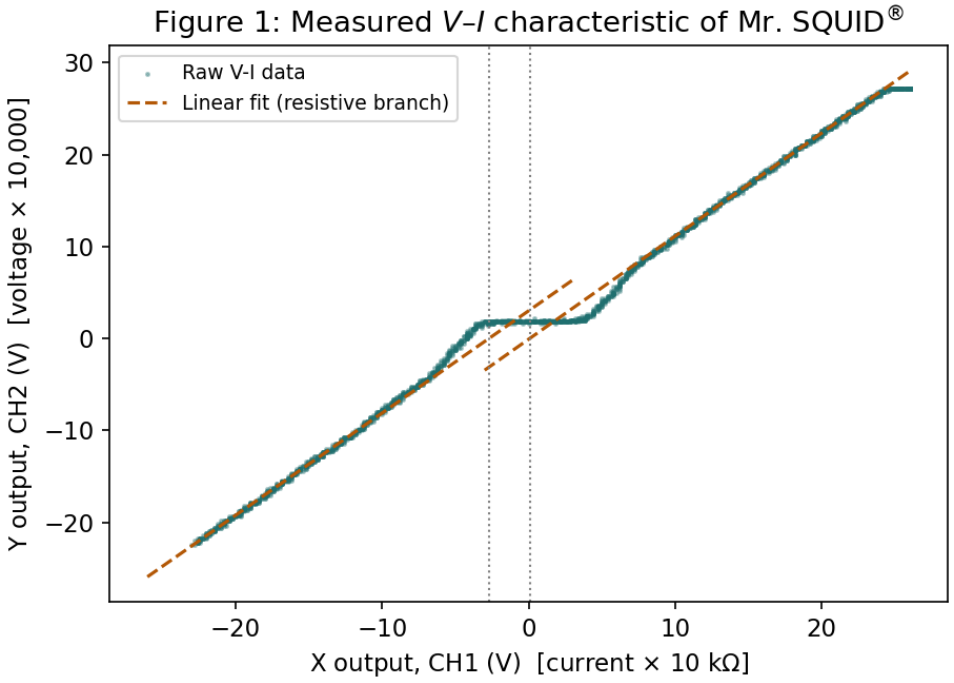


Figure 2: Measured V–I characteristic (bin-averaged). Dashed lines are linear fits to the resistive branches; their $V_Y = 0$ intercepts (vertical dotted lines) bracket the critical-current plateau of width $4I_c$.

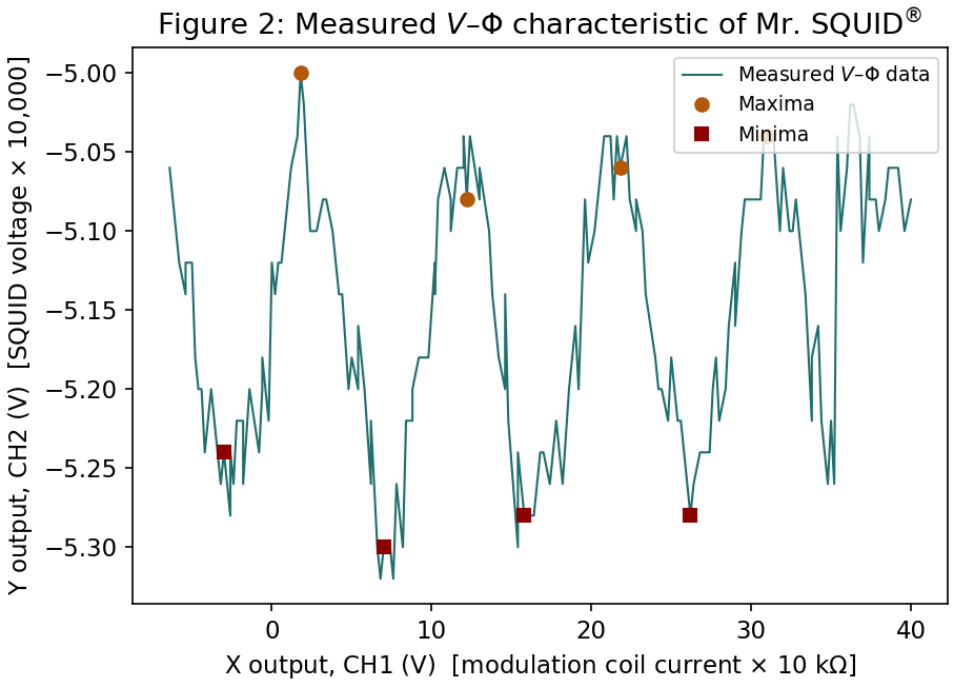


Figure 3: Measured V–Φ characteristic for one monotonic leg of the modulation-coil sweep. Circles and squares mark the four peaks and troughs used to determine the flux period and ΔV.

7 Discussion

The two β_L values obtained from the zero-temperature expressions (4.03 from the inductance and 16.18 from modulation depth) disagree by roughly a factor of four. Applying the thermally corrected expression brings the modulation-depth estimate down to $\beta_L^{(3\text{-}5)} = 8.70$. The correction factor $(1 - 3.57\sqrt{k_B T L/\Phi_0^2}) = 0.564$ accounts for essentially all of the qualitative disagreement, cutting it to a factor of $\sim$2. This confirms quantitatively that thermal rounding of the junctions is the dominant cause of the gap between Eqn. 3-1 and Eqn. 3-2. However, it does not fully close the gap, possibly due to the following reasons: Both $I_c R_N = 310$ µV and $2I_c = 139$ µA measured for this probe are at or above the high end of the typical ranges (10–100 µV and 5–150 µA respectively), and the nominal $L = 60$ pH used in equations 3-1 and 3-5 is a manufacturer average rather than a value measured for this specific chip.

The dominant systematic uncertainty is the linear extrapolation used to locate the knee of the V–I curve; because the curve is thermally rounded near the knee, the fit range chosen for the resistive branches has a small but non-negligible effect on the extracted I_c and R_N, and hence on $I_c R_N$ and $\beta_L^{(3\text{-}2)}$. The V–Φ period was reproducible to about 7% (standard deviation of six independent period estimates), which propagates directly into the uncertainty on ΔI and M. A repeat of this experiment could reduce the dominant uncertainty by averaging several complete V–Φ periods using a longer record and by adjusting the flux-bias control to maximize the plateau width before each V–I acquisition.

8 Conclusion

The V–I and V–Φ characteristics of a 77 K YBCO dc SQUID were measured with a digital oscilloscope. The junction critical current was $I_c = 69.46$ µA, the normal-state resistance $R_N = 4.469$ Ω, and the characteristic voltage $I_c R_N = 310.4$ µV; the peak-to-peak flux modulation was $\Delta V = 23.0$ µV. The modulation parameter β_L was found to be 4.03 from the nominal loop inductance (Eqn. 3-1), 16.18 from the zero-temperature modulation-depth expression (Eqn. 3-2), and 8.70 once the thermal correction of Eqn. 3-5 is applied at $T = 77$ K. The thermal correction accounts for most, but not all, of the disagreement between Eqn. 3-1 and Eqn. 3-2, due to thermal-noise effects on high-T_c SQUIDs at 77 K. All stated aims of the experiment were achieved.

Precautions

- Do not remove covers/panels or attempt repairs beyond those specified in the manual; return the instrument to STAR Cryoelectronics for service.
- Do not operate the instrument near flammable gases or fumes.
- Liquid nitrogen causes severe burns on skin contact – wear eye protection and gloves, and keep an experienced person on hand for all cryogenic work.
- After removing the probe from the dewar, do not touch the cold end until frost/condensation has fully disappeared.
- Cool the SQUID with minimal nearby magnetic field (power down nearby computers/radios) to avoid flux trapping, and wait a few minutes after cooldown before increasing the sweep amplitude.

A Datasets of the experiment

Oscilloscope waveform files can be found in this directory.

B Plotting and fitting

The Google Colab file made to plot the current–voltage (V–I) and voltage–flux (V–Φ) characteristics can be found here.